

1. A particle A has mass 4 kg and a particle B has mass 2 kg .

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u\text{ ms}^{-1}$ and the speed of B is $3u\text{ ms}^{-1}$

Immediately after the collision, the speed of B is $2u\text{ ms}^{-1}$

The direction of motion of B is reversed by the collision.

- (a) Find, in terms of u , the speed of A immediately after the collision.

(3)

- (b) State the direction of motion of A immediately after the collision.

(1)

- (c) Find, in terms of u , the magnitude of the impulse received by B in the collision.
State the units of your answer.

(3)

[In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1 \text{ N}$, $\mathbf{F}_2 \text{ N}$ and $\mathbf{F}_3 \text{ N}$ where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$

$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P .

Given that P remains in equilibrium,

- (a) find $\mathbf{F}_3$ in terms of c , $\mathbf{i}$ and $\mathbf{j}$.

(2)

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

- (b) find the size of the angle between the direction of $\mathbf{i}$ and the direction of the resultant force acting on P .

(4)

The mass of P is $m \text{ kg}$.

Given that the magnitude of the acceleration of P is 8.5 ms^{-2}

- (c) find the value of m .

(4)

3. Two students observe a book of mass 0.2 kg fall vertically from rest from a shelf that is 1.5 m above the floor.

Student A suggests that the book is modelled as a particle falling freely under gravity.

- (a) Use student A 's model to find the time taken for the book to reach the floor.

(3)

Student B suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude R newtons.

Given that the time taken for the book to reach the floor is 0.6 seconds,

- (b) use student B 's model to find the value of R

(5)

4.

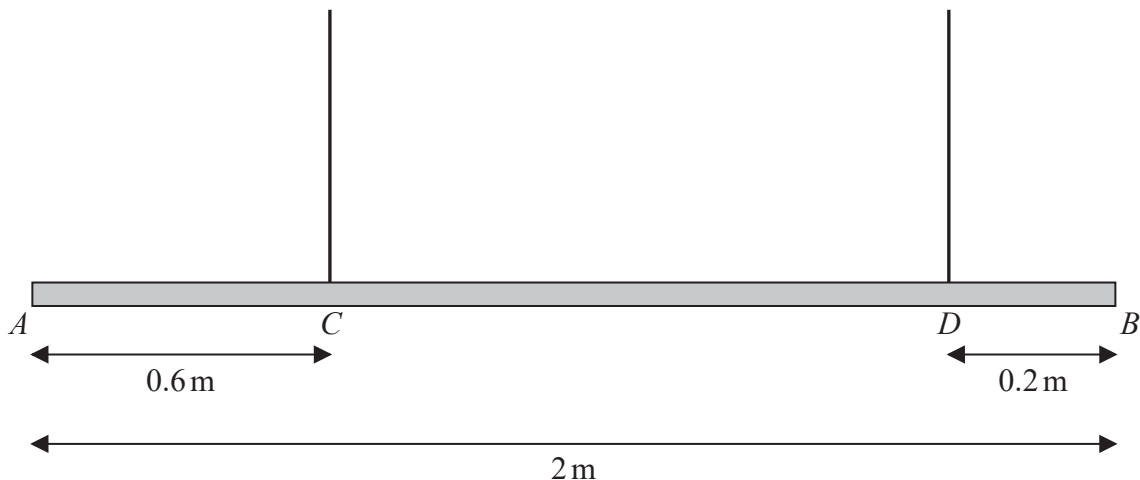


Figure 1

Figure 1 shows a beam AB , of mass m kg and length 2 m, suspended by two light vertical ropes.

The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m

The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

- (a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D , use the model to find the exact value of p .

(7)

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A .

The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

- (b) Using the model, find the exact value of q

(4)

- (c) State how you have used the modelling assumption that the beam is uniform.

(1)

5.

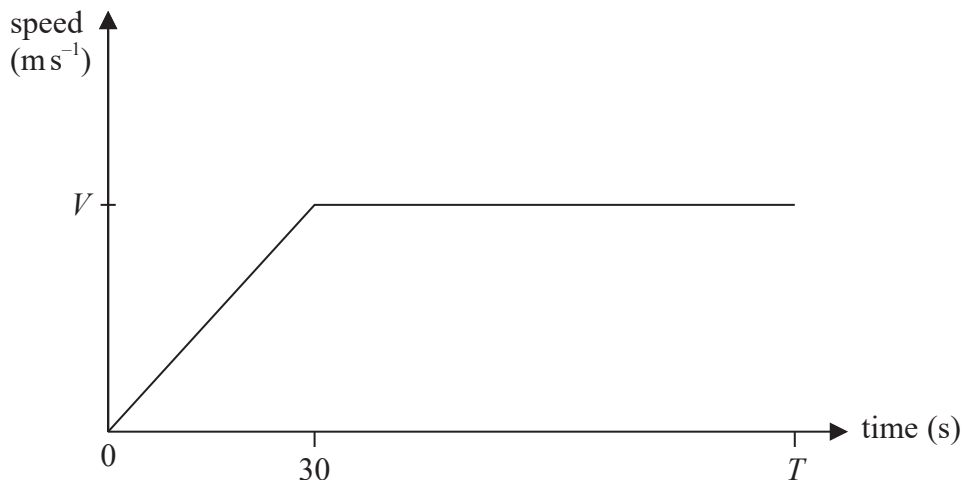


Figure 2

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.

At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V\text{ms}^{-1}$

The car then travels at a constant speed of $V\text{ms}^{-1}$ until time $t = T$ seconds.

- (a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

A motorbike also travels along the same road.

- The motorbike starts from rest at time $t = 10\text{ s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V\text{ms}^{-1}$ at time T seconds

- (b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

[If you need to redraw your sketch, there is a copy of Figure 2 on page 15.]

(2)

- (c) Show that the constant speed of the motorbike is $\frac{4V}{3}\text{ms}^{-1}$

(2)

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

- (d) Find the value of T

(5)

6.

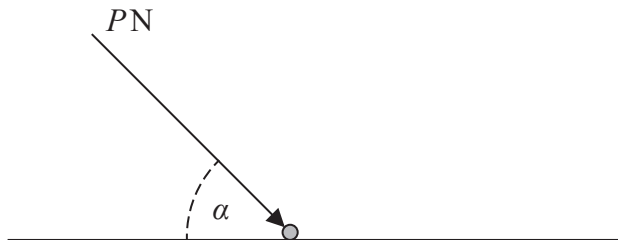


Figure 3

A particle of weight W newtons lies at rest on a rough horizontal surface, as shown in Figure 3.

A force of magnitude P newtons is applied to the particle.

The force acts at an angle α to the horizontal, where $\tan \alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is $\frac{1}{4}$

Given that the particle does not move, show that

$$P \leq \frac{5W}{8}$$

(7)

7.

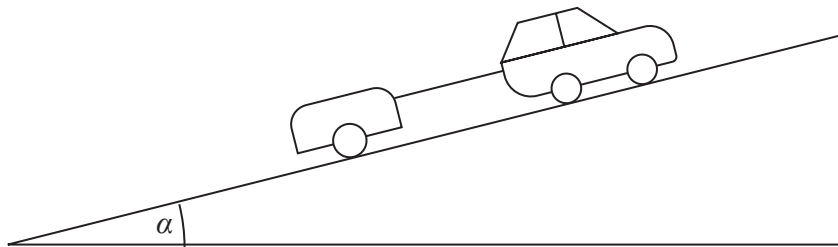


Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration 0.75 m s^{-2}

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude $2R$ newtons
- the **trailer** is modelled as a constant force of magnitude R newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of R is 60

(4)

(b) find the tension in the tow bar.

(3)

When the car and trailer are moving at a speed of 12 m s^{-1} , the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

8.

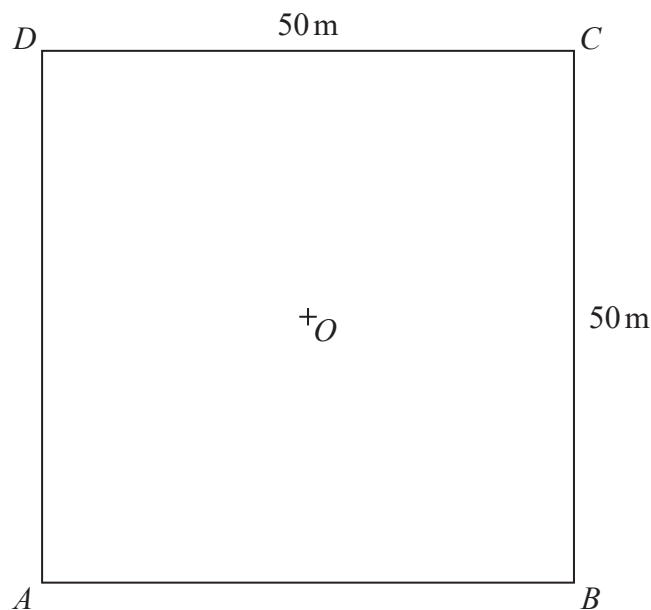


Figure 5

A square floor space $ABCD$, with centre O , is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.

The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to O .

A small robot R is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, R is at the point with position vector $(-2\mathbf{i} + \mathbf{j})\text{m}$
- At time $t = 11\text{ s}$, R is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})\text{m}$
- At time t seconds, the position vector of R is $\mathbf{r}$ metres

(a) Find, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{r}$

(3)

A second robot S is at the point C .

- At time $t = 0$, S leaves C and moves with constant velocity $(-\mathbf{i} - \mathbf{j})\text{ms}^{-1}$
- At time t seconds, the position vector of S is $\mathbf{s}$ metres

(b) Write down, in terms of t , $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{s}$

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}] \text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)